PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Skill decay following Basic Life Support training – a systematic review protocol
AUTHORS	Stanley, Benjamin; Burton, Thomas; Percival, Harriet; Beesley, Emily; Coffin, Nicholas; Hulme, Jonathan; Owen, Andrew; Alderman, Joseph

VERSION 1 – REVIEW

REVIEWER	Andrews, Tom University College Cork National University of Ireland, Nursing and Midwifery
REVIEW RETURNED	29-Jun-2021

GENERAL COMMENTS	Thank you for the opportunity of reviewing this manuscript. It is very well written , with a clear and appropriate outline for undertaking a systematic review.

REVIEWER	Chen, Chien-Yu Taipei Medical University Hospital, Taipei, Taiwan, Department of Anesthesiology
REVIEW RETURNED	01-Jul-2021

GENERAL COMMENTS	1. The topic of this systematic review is crucial to future policymaking for BSL education design. 2. The research questions were all proposed in a sound manner. 3. The searching strategies in Appendix 1 by using three “AND” is inappropriate and might limit the outcome. The row 2,3,4 should be combined by using OR rather than AND. Only one AND should be applied in between row 1 and row 2/3/4 so that the searching is comprehensive enough to answer the questions proposed. 4. To identify unpublished studies, clinicaltrials.gov should be included. 5. The outcomes assessed and observed were not clearly specified. Please define whether knowledge, skills, or behaviors are aimed to be reviewed in this study.

REVIEWER	Hultin, Magnus Umeå University, Department of Surgical and Perioperative Sciences, Anaesthesiology and Intensive Care
REVIEW RETURNED	23-Jul-2021

GENERAL COMMENTS	Thank you for the opportunity to review your protocol for a systematic review of the decay of skills after Basic Life Support training. It is an interesting and important topic and the results of the analysis will be useful for designing training programs for

	undergraduates, postgraduates and for continuous professional development. The protocol includes the key elements specified in PRISMA-P guidelines. However, there are a number of issues that needs to be addressed Major points 1. The research questions are properly described and the outlined protocol is sound. However, the manuscript describing the protocol is in my opinion still needs further development and is not yet suitable for publication. E.g. there is no Discussion and some of the paragraphs are just a list of words. These may either be moved to a table and referenced in the text or converted to a coherent paragraph. See for example https://systematicreviewsjournal.biomedcentral.com/articles/10.1186/s13643-021-01761-y or https://bmjopen.bmj.com/content/11/7/e046163. 2. A more traditional use of headings would be beneficial for the reader. I suggest that you use the major headings Introduction, Methods and analysis, and Discussion, and put the headings currently used in the manuscript as subheadings Minor points 3. Abstract/Introduction: Suggests that you specify which national curriculum that has been changed to include BLS? 4. Inclusion and exclusion criterias: Perhaps move the criteria to a table that is referenced in the text? 5. Search strategy: The search languages are different for the different databases listed in the protocol. Only the search string for Medline is listed. How will the search-strings for the other databases be developed and reported? 6. Data sources: Why include both Pubmed and Medline? Isn't it the same database (medline) that is being searched using Pubmed as the front-end? Which relevant journals and which preprint servers will be included in the search? Perhaps convert the list into a table that is referenced into the paragraph search strategy. 7. Database and statistics. Will a commercially available database or tool be used to collect and analyze the data or will a new database be developed specifically designed for this purpose? Which statistical analyses will be performed on the collected data and using which tools? 8. Author statement: It is unclear which contributions to the manuscript that most of the listed authors actually have made. Please be more specific. Appendix 1-4 9. Do you have a mechanism in place for making adjustments to the search strings and in the forms or are they final? Have you already piloted the screening/extraction forms? 10. Appendix 1: Why is there double OR in [TIAB] search strategy no 1 and no 2? 11. Appendix 4: Data-extraction: The quality of each assessment and reassessment depends on how the raters were trained. I would suggest to also extract information on how raters were trained and if any potential quality of assessment statistics were published in the screened studies.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Tom Andrews, University College Cork National University of Ireland
Comments to the Author:
Thank you for the opportunity of reviewing this manuscript. It is very well written , with a clear and appropriate outline for undertaking a systematic review.

Response: Many thanks for your review and kind comments

Reviewer: 2
Dr. Chien-Yu Chen, Taipei Medical University Hospital, Taipei, Taiwan
Comments to the Author:
1. The topic of this systematic review is crucial to future policymaking for BSL education design.
2. The research questions were all proposed in a sound manner.

Response: Many thanks for your review and recognition of the importance of relevance of this systematic review

3. The searching strategies in Appendix 1 by using three “AND” is inappropriate and might limit the outcome. The row 2,3,4 should be combined by using OR rather than AND. Only one AND should be applied in between row 1 and row 2/3/4 so that the searching is comprehensive enough to answer the questions proposed.

Response: Many thanks for your comments on the searching strategy. This has been adapted to demonstrate a more comprehensive strategy with more search terms now combined using OR rather than AND

4. To identify unpublished studies, clinicaltrials.gov should be included.

Response: Many thanks for this helpful suggestion. Clinicaltrials.gov has now been added as data source for unpublished studies

5. The outcomes assessed and observed were not clearly specified. Please define whether knowledge, skills, or behaviors are aimed to be reviewed in this study.

Response: Many thanks for raising this ambiguity. The outcomes have now been adapted to clarify that that it is practical BLS skills being assessed as the outcome for included studies

Reviewer: 3
Dr. Magnus Hultin, Umeå University
Comments to the Author:
Thank you for the opportunity to review your protocol for a systematic review of the decay of skills after Basic Life Support training. It is an interesting and important topic and the results of the analysis will be useful for designing training programs for undergraduates, postgraduates and for continuous professional development.

Response: Many thanks for your review and the recognition of the importance and relevance of this work

The protocol includes the key elements specified in PRISMA-P guidelines. However, there are a number of issues that needs to be addressed

Major points
1. The research questions are properly described and the outlined protocol is sound. However, the manuscript describing the protocol is in my opinion still needs further development and is not yet suitable for publication. E.g. there is no Discussion and some of the paragraphs are just a list of words. These may either be moved to a table and referenced in the text or converted to a coherent paragraph. See for example
https://systematicreviewsjournal.biomedcentral.com/articles/10.1186/s13643-021-01761-y or
https://bmjopen.bmj.com/content/11/7/e046163.
2. A more traditional use of headings would be beneficial for the reader. I suggest that you use the major headings Introduction, Methods and analysis, and Discussion, and put the headings currently used in the manuscript as subheadings

Response: Many thanks for your comments and for providing the above examples. The manuscript has been significantly adapted to improve the suitability for publication. A discussion section has now been added which discusses the importance of the review, potential limitations and how these will be overcome. The overall structure of the manuscript has also been adapted to reflect the traditional 'Introduction, Methods, Discussion' structure, with the original subheadings adapted into this. The list of data sources has now been incorporated into the 'search strategy' section as prose to improve flow and readability.

Minor points
3. Abstract/Introduction: Suggests that you specify which national curriculum that has been changed to include BLS?

Response: Many thanks for raising this. Both the abstract and introduction have been adapted to clarify that it is the English national curriculum into which BLS teaching has now been incorporated
4. Inclusion and exclusion criterias: Perhaps move the criteria to a table that is referenced in the text?

Response: Many thanks for this helpful suggestion. The inclusion and exclusion Criteria have been converted into a table format and appropriately referenced in text
5. Search strategy: The search languages are different for the different databases listed in the protocol. Only the search string for Medline is listed. How will the search-strings for the other databases be developed and reported?

Response: Many thanks for raising this. The search strategy included as an appendix has been devised alongside the library team at the University of Birmingham. Whilst this is specific to Medline, minor adaptations will be made for the searching of other databases. Final search strategies for all databases will be included as appendices in the final review, as now outlined in the protocol manuscript
6. Data sources: Why include both Pubmed and Medline? Isn't it the same database (medline) that is being searched using Pubmed as the front-end? Which relevant journals and which preprint servers will be included in the search? Perhaps convert the list into a table that is referenced into the paragraph search strategy.

Response: Many thanks for pointing out the duplication of Pubmed and Medline. Pubmed has now been removed from the list to rectify this. 'Resuscitation' and the 'Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine' have been added as specific relevant journals which will be searched given their importance in the field. medRxiv has been added as a pre-print server to be searched with Clinicaltrials.gov also added as a data source for unpublished studies

7. Database and statistics. Will a commercially available database or tool be used to collect and analyze the data or will a new database be developed specifically designed for this purpose? Which statistical analyses will be performed on the collected data and using which tools?

Response: Many thanks for the comment. The manuscript has now been clarified to demonstrate that a bespoke database will be produced specifically for this project, using commercially available database software such as Microsoft access. 'R' will be used to perform statistical analysis; this has been added to the manuscript. Exact statistical analysis performed will be dependent on the data extracted from primary literature. As outlined in the manuscript, meta-analysis will be performed provided the level of heterogeneity (as assessed by I2) is not too great

8. Author statement: It is unclear which contributions to the manuscript that most of the listed authors actually have made. Please be more specific.

Response: Many thanks for raising this. The author contributions, both within the manuscript and within ScholarOne has now been adapted to clarify the specific contributions made by the author Appendix 1-4

9. Do you have a mechanism in place for making adjustments to the search strings and in the forms or are they final? Have you already piloted the screening/extraction forms?

Response: Many thanks for your comments and questions. The search strings and extraction forms have been piloted and found to be functional. As these will be used in an electronic format, changes can be made and easily tracked. Changes will only be made following consultation with senior reviewer. Any changes will be outlined in the final review. This information has been updated in the protocol manuscript

10. Appendix 1: Why is there double OR in [TIAB] search strategy no 1 and no 2?

Response: Many thanks for alerting to this. The repeated OR was a formatting error and has been removed

11. Appendix 4: Data-extraction: The quality of each assessment and reassessment depends on how the raters were trained. I would suggest to also extract information on how raters were trained and if any potential quality of assessment statistics were published in the screened studies.

Response: Many thanks for this helpful suggestion. Extraction of information relating to level of training of BLS assessors has been added to appendix

VERSION 2 – REVIEW

REVIEWER	Chen, Chien-Yu Taipei Medical University Hospital, Taipei, Taiwan, Department of Anesthesiology
REVIEW RETURNED	27-Sep-2021

GENERAL COMMENTS	The authors have replied to the questions and revised accordingly.

REVIEWER	Hultin, Magnus Umeå University, Department of Surgical and Perioperative Sciences, Anaesthesiology and Intensive Care
REVIEW RETURNED	19-Sep-2021

GENERAL COMMENTS	Thank you for adjusting the protocol according to the suggestions. Best wishes with the study.